

Social Promotion & Tracking + Customer/Scheduled Email

Capability audit, locked decisions, and implementation plan (Wave 6)

2026-07-16 · litemall microservices platform · status: **decisions locked, worktree blocks issued**

1. Executive summary

Questions asked: (a) Can the system support our social-media pages — posting specific goods for promotion and tracking activity/events around our products there? (b) Can it schedule email sending — messages and confirmations — to customers or third parties?

Verdict: neither exists as a working feature today, but both are unusually cheap to build here because the promotion service already ships a complete, tested, *disabled* marketing-stack integration layer: a **Matomo** analytics reader feeding the RFM targeting engine, and a **Mautic** marketing-automation writer that pushes campaign audiences out as segments. Neither backend is deployed, the storefront carries no tracker, and email today can reach exactly one fixed operator mailbox. The plan below turns the existing seams on, adds the genuinely missing pieces (platform posting adapters, a customer mail sender with a scheduled outbox), and closes the loop.

Prerequisites only you can provide (they gate the timeline): a Meta business app with Facebook-Page + Instagram-business tokens; TikTok developer / Content-Posting-API approval (longest lead time — apply first); real SMTP credentials for production email (development uses a MailHog container).

2. Audit — what exists today (verified in code and dev DB)

Asset	State	Detail
Social platform APIs (Meta/X/TikTok)	ABSENT	No clients, no OAuth apps, no share buttons, no post composer. SPA "social" grep hits are CoreUI admin-template leftovers.
Storefront analytics	ABSENT	No tracker snippet, no UTM handling in the customer SPA — inbound social traffic is invisible today.
Matomo integration (analytics IN)	CODED, OFF, NOT DEPLOYED	promotion/infrastructure/acl/matomo/ : Feign reporting client + statistics adapter feeding CustomerStatisticsProvider → RFM targeting. Source switched by <code>litemall.promotion.stats.source</code> . No Matomo container exists.

Mautic integration (campaigns OUT)	CODED, OFF, NOT DEPLOYED	<code>acl/mautic/</code> : segment-create, contact-upsert, add-to-segment; fired by <code>PromotionTargetedEvent</code> AFTER_COMMIT. Contacts are keyed by user id only — no email in the payload (emails did not exist pre-Wave-4). No Mautic container exists.
Campaign vertical	LIVE	<code>litemall_promotion_campaign</code> : RFM score minimums, segments, target goods, start/end window, audience & budget caps.
Customer email	MISSING	Core <code>NotifyService.notifyMail</code> sends to ONE fixed operator address (<code>notify.mail.enable:false</code> , placeholder creds); only caller = "New order notification". Wave-4's password-reset mail shipped as a NO-OP logging port. No SMTP anywhere. User emails exist only for post-Wave-4 registrations (nullable).
Scheduling machinery	PATTERN EXISTS	No mail scheduler, but <code>litemall_event_outbox</code> (V25) provides the outbox-plus-sweep pattern to copy; order runs several <code>@Scheduled</code> sweeps already.
TikTok video sources	AVAILABLE	Wave-3 shipped <code>/srv/goods/videos</code> + CJ video flags — partial catalog coverage; the composer must grey TikTok out for video-less goods.
Deal lifecycle (auto-post trigger)	LIVE	Flash-deal price-swap lifecycle (deals wave, V38) in goods-management — activation is the natural opt-in auto-post hook.

3. Locked decisions (user-approved 2026-07-16)

Decision	Choice	Rationale
Posting platforms (v1)	Meta (Facebook Page + Instagram) AND TikTok	One Graph-API adapter covers FB+IG; TikTok is video-first and rides the existing CJ product videos.
Activity tracking	Deploy Matomo + storefront tracker snippet	The reporting adapter already exists — deploying the server and injecting the tracker turns on campaign reports, product funnels, referrer breakdowns, and real RFM inputs with nearly zero new report code.
Post trigger	Manual admin composer + OPT-IN auto-post on deal activation	Control by default; the deals lifecycle provides the automation hook when wanted.
Email capability	In-house transactional (port + SMTP + outbox with <code>send_at</code>) + Mautic for scheduled/drip campaigns	Confirmations must not depend on an external marketing box; campaign scheduling, sequences, and unsubscribe come free from Mautic's UI over the already-coded segment push.

4. Implementation plan

Phase A — Tracking foundation (Matomo) · worktrees: promotion (infra) + gateway-api

1. `docker-compose/docker-compose.marketing.yml` : Matomo + MariaDB, Mautic + DB, MailHog (dev SMTP) — env-configured, own volumes, opt-in start (not in the default local set).
2. Storefront SPA: Matomo tracker (config-driven, absent ⇒ exact current behavior), SPA route-change page views, product-detail views carrying a goods-id custom dimension; honor Do-Not-Track.
3. Shared UTM convention `utm_source=facebook|instagram|tiktok & utm_medium=social & utm_campaign=<slug>` in a link-builder reused by social captions and the affiliate links page (compose with `?invite=`).
4. Config-only: flip `litemall.promotion.stats.source` to composite once Matomo has data — the adapter is already written.

Phase B — Social posting vertical · worktrees: promotion (owns) + gateway-admin (UI)

1. Migration `litemall_social_post` (goods, platform meta_fb|meta_ig|tiktok, caption, media/link URLs, status draft|posted|failed, external_post_id, error, posted_by) + litemall-db mapper trio.
2. `SocialPublishPort` + ACL adapters (`acl/meta` — Page feed/photo post, IG media-container→publish; `acl/tiktok` — Content Posting, video-first). Disabled by default; every failure degrades to a `failed` row + log, never a 5xx — the Mautic adapter's conventions exactly.
3. Admin endpoints: compose-preview (caption template with price/deal price/UTM share link + candidate media), post (loops selected platforms, one ledger row each), list, retry.
4. Opt-in auto-post on flash-deal activation (`auto-post-deals:false` default), v1 via a deal-table poll inside promotion (no cross-module edit), rows marked `posted_by='auto'` .
5. Admin SPA: "Promote" composer dialog (caption editor, media picker, platform checkboxes with the TikTok video gate, preview, per-platform toasts) + Social posts list (status chips, retry, link-out).

Phase C — Transactional email with real scheduling · worktrees: order (owns core edit) + gateway-api (reset adapter) + gateway-admin (pages)

1. Migration `litemall_mail_outbox` (recipient, subject, body, template_key, status pending|sent|failed, attempts, `send_at` — native scheduled sending, last_error) — the

V25 event-outbox pattern.

2. litemall-core `mail` package: `CustomerMailSender` port + SMTP adapter (`litemall.customer-mail.*`, `enabled:false`, creds via env — core config-precedence trap) + English templates: order-confirmation, shipped(+tracking), refund-approved, pickup-code, password-reset.
3. Order listeners (`AFTER_COMMIT`, never block payment) on paid/shipped/refund/write-off enqueue outbox rows (user email NULL ⇒ silent skip); a `@Scheduled` sweep sends `pending WHERE send_at <= now` with attempt caps; admin mail list + resend.
4. gateway-api: the Wave-4 no-op `ResetMailSender` gets a real SMTP adapter on the same config keys.

Phase D — Scheduled campaign email (Mautic) · worktrees: promotion + gateway-admin (link-out)

1. Mautic container (Phase A file) + SMTP config (same creds as Phase C).
2. Small adapter edit: contact upsert gains the user's EMAIL + nickname (today it sends only the litemall user id).
3. Flow (config + docs, not code): admin builds a campaign (RFM/segments/goods/window) → audience lands in a Mautic segment automatically → scheduled sends, drip sequences, and unsubscribe pages are built in Mautic's UI. One-off scheduled third-party mail = a Phase-C outbox row with a future `send_at`.

5. Verification highlights

- A visit through a `utm_source=facebook` link appears in Matomo's campaign report; product views carry the goods dimension; RFM evaluation consumes Matomo rows; tracker unconfigured ⇒ zero behavior change.
- Composer posts a goods to FB + IG (row → posted + external id); TikTok greyed without video and posts with one; dead creds ⇒ failed row + toast, never a 5xx; deal activation auto-posts only when enabled; the posted links' clicks show up in Matomo — the full loop.
- Paying as an email-bearing user delivers a confirmation (MailHog in dev); email-less user ⇒ silent skip; SMTP down ⇒ payment still succeeds, row failed, resend works; a future `send_at` row sends only after that time — the scheduled-send proof; password-reset mail delivers when enabled; all defaults-off ⇒ byte-identical system behavior.
- Campaign evaluation fills a Mautic segment with real emails; a scheduled Mautic email lands for segment members; unsubscribe honored.

6. Sequencing, risks, status

- **Order:** A (small foundation) and C (email) are independent — start together with the platform app registrations; B lands once Meta/TikTok credentials exist (adapters are built and testable disabled before that); D after C's SMTP.
- **Risks:** TikTok approval lead time + video coverage; Meta long-lived page-token renewal (documented); GDPR/unsubscribe for campaign mail (Mautic-native; transactional mail keeps a kill switch); pre-Wave-4 users have no email address.
- **Status:** decisions locked and converted into CLAUDE.md Wave-6 worktree blocks (promotion, order, gateway-api, gateway-admin) on 2026-07-16; plan also persisted to session memory (`social-email-marketing-plan`). Wave-5 affiliate merged the same day — the affiliate links page and this plan share the UTM/invite link builder.

Sources: code audit of litemall-promotion-service `infrastructure/acl/{matomo,mautic,nutch}` + `docs/phase3-marketing-stack-integration.md`, litemall-core `notify` package + `application-core.yml`, gateway-api auth/ResetMailSender, docker-compose inventory, storefront SPA grep for trackers/share/UTM; dev-DB DESCRIBE of litemall_promotion_campaign; deals-wave and Wave-3 video endpoints per merged history.